Supply Chain Late Delivery Risk Prediction using Machine Learning

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Executive Summary

This project predicts whether a supply chain order is at risk of late delivery using supervised machine learning. The workflow included data cleaning, exploratory data analysis, feature engineering, model training, evaluation, deployment with Streamlit, and version control with GitHub.

Problem Statement

Organizations need early warning of late deliveries so they can take corrective action before customers are affected. The goal is to predict late delivery risk using information available at decision time.

Dataset

DataCo Supply Chain Dataset with approximately 180,519 records. After preprocessing, 46 features were used for modelling. Categorical variables were encoded and missing values handled.

EDA Findings

The dataset contained numeric and categorical features, some missing values, and class imbalance. Visual exploration guided preprocessing and feature selection.

Preprocessing

Encoded categorical columns, handled missing values, split the data into training and testing sets, and prepared the feature matrix for machine learning.

Models

Three models were evaluated: Logistic Regression, Decision Tree, and Random Forest.

Results

Logistic Regression achieved about 55% accuracy. Decision Tree and Random Forest achieved 100% accuracy on the prepared dataset. Such unusually high performance suggests potential data leakage, so production use would require removing post-decision features and retraining.

Deployment

The trained Random Forest model was saved as a .pkl file and loaded in a Streamlit application for demonstration.

GitHub

The complete project, notebooks, model, documentation, and application were uploaded to a GitHub repository.

Limitations

Some features may contain information unavailable at prediction time, causing leakage. Future work should remove such features, tune hyperparameters, and validate on unseen temporal data.

Conclusion

The project demonstrates an end-to-end machine learning workflow from data preprocessing to deployment and documentation.